FOUR EQUIVALENT FORMS OF THE FOURIER ENTROPY-INFLUENCE CONJECTURE

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ABSTRACT. Let f be a Boolean function with spectral distribution $\mu(S) = \hat{f}(S)^2$, and let $\gamma(T) = \sum_{U \supseteq T} \mu(U)$ be the generalized influence of a set T. The Fourier entropy splits as $H(f) = A + B$ with $A = \mathbb{E}_\mu \log_2(1/\gamma(S))$ and $B = \mathbb{E}_\mu \log_2(\gamma(S)/\mu(S)) \geq 0$. Since derivatives of Boolean functions take values in a lattice, $A \leq H(f)/2 + I(f) - \mathrm{Var}(f)$, and so the Fourier entropy-influence conjecture is equivalent to $B \leq C\, I(f)$. We give three further equivalent forms: a bound on the cross-entropy of the spectral distribution against the noisy generalized influences $\|T_{\sqrt{t}}\, \partial_T f\|_2^2$ at a fixed $t < 1$; a bound on the integral over $t \in [\frac{1}{2}, 1]$ of the expected number of elements a noise-tilted posterior adds to the spectral sample; and a bound of $O(1)$ bits on the log-loss of predicting a uniformly chosen element of the size-biased spectral sample from the others, which also has a variational form over all predictors. Form (c) gives the unconditional bound $H(f) \leq 4I(f) - 2\mathrm{Var}(f) + 2M(f)$, where $M(f)$ is the expected number of such added elements at $t = 1$.

1. INTRODUCTION

Let $f : \{-1,1\}^n \to \{-1,1\}$ have Fourier expansion $f = \sum_S \hat{f}(S)\chi_S$. By Parseval $\mu(S) = \hat{f}(S)^2$ is a probability distribution on subsets of $[n]$, and we write U for a random set with law μ. Put

$$H(f) = \sum_S \mu(S) \log_2 \frac{1}{\mu(S)}, \quad \mathrm{Inf}_i(f) = \sum_{S \ni i} \mu(S), \quad I(f) = \sum_S |S| \mu(S), \quad \mathrm{Var}(f) = 1 - \mu(\emptyset).$$

The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [1] asks for a universal C with $H(f) \leq C\, I(f)$ for all Boolean f. It is known for symmetric functions and read-once decision trees [9], it is preserved under suitable compositions [8], there are bounds in terms of decision-tree and communication complexity measures [2, 13], and it holds for linear threshold functions [11]. For every Boolean f, Han [4] proved $H(f) \leq (2 + 3/\ln 2)\, I(f) + \sum_i \mathrm{Inf}_i(f) \log_2(1/\mathrm{Inf}_i(f))$, and Li and Han [6] lowered the constant in front of $I(f)$ to the sharp value $1/\ln 2$. In [10] the conjecture was shown to be equivalent to a one-coordinate inequality, and to the bound $\sum_i E_i(f) \leq C\, I(f)$ on the erasure entropies (in the sense of Verdú and Weissman [12])

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big).$$

The one-coordinate route goes back to Han [4, Remarks 3.1–3.2], and the splitting of the entropy drop at one coordinate into E_i and a Jensen–Shannon term is equation (32) of González, MacManus and Pereyra [3]. Here we give four more equivalent forms. All of them use the derivatives

$$\partial_T f = \sum_{U \supseteq T} \hat{f}(U) \chi_{U \setminus T}, \qquad \gamma_t(T) = \|T_{\sqrt{t}}\, \partial_T f\|_2^2 = \sum_{U \supseteq T} \mu(U)\, t^{|U \setminus T|} \quad (0 \leq t \leq 1),$$

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where T_ρ is the noise operator. For $t = 1$ we write $\gamma(T) = \gamma_1(T) = \|\partial_T f\|_2^2$; this is the generalized influence $I_T[f]$ of Kelman, Kindler, Lifshitz, Minzer and Safra [5]. Note $\gamma_0 = \mu$, $\gamma(\emptyset) = 1$ and $\mu \le \gamma_t \le \gamma \le 1$. Sums over S below run over the support of μ. Put

$$A = \sum_S \mu(S) \log_2 \frac{1}{\gamma(S)}, \qquad B = \sum_S \mu(S) \log_2 \frac{\gamma(S)}{\mu(S)}, \qquad H(f) = A + B,$$

and for $0 < t < 1$ let $A_t = \sum_S \mu(S) \log_2(1/\gamma_t(S))$. Let

$$m_t(T) = \frac{\sum_{U \supseteq T} |U \setminus T|\, \mu(U)\, t^{|U \setminus T|}}{\gamma_t(T)}, \qquad M_t(f) = \sum_S \mu(S)\, m_t(S),$$

so that $m_t(T)$ is the mean of $|U \setminus T|$ when $U \supseteq T$ is drawn with probability proportional to $\mu(U)t^{|U\setminus T|}$, and write $M(f) = M_1(f)$. Finally let Q be the law on pairs (T, i) with $i \notin T$ given by $Q(T, i) = \mu(T \cup \{i\})/I(f)$; it is the law of $(U \setminus \{i\}, i)$ when U is drawn size-biased and i is a uniform element of U. Put

$$L(f) = \mathbb{E}_{(T,i)\sim Q} \log_2 \frac{\mu(T) + \mu(T \cup \{i\})}{\mu(T \cup \{i\})},$$

the log-loss in bits of predicting that $i \in U$ from $U \setminus \{i\}$.

Theorem 1. *For every Boolean function f and every $0 < t < 1$,*

(a) *$H(f) \le 2B + 2I(f) - 2\mathrm{Var}(f)$ and $0 \le B \le H(f)$;*
(b) *$A_t \le H(f) \le A_t + I(f) \log_2 \frac{1}{1-t}$;*
(c) *$B = \frac{1}{\ln 2} \int_0^1 M_s(f)\, \frac{ds}{s}$, the function $s \mapsto m_s(T)$ is nondecreasing for every T, and*

$$H(f) \le 4I(f) - 2\mathrm{Var}(f) + \frac{2}{\ln 2} \int_{1/2}^1 M_s(f)\, \frac{ds}{s} \le 4I(f) - 2\mathrm{Var}(f) + 2M(f);$$

(d) *$I(f)\, L(f) \le \sum_i E_i(f) \le I(f)\, L(f) + I(f)/\ln 2$.*

Consequently, with [10, Corollary 5] *for the last item, the following are equivalent: the FEI conjecture; $B \le C\, I(f)$ for all f; $A_t \le C\, I(f)$ for all f (for one, or every, fixed $t \in (0, 1)$); $\int_{1/2}^1 M_s(f)\, ds \le C\, I(f)$ for all f; and $L(f) \le C$ for all f.*

Each form isolates a different piece of the difficulty. Form (a) says that the part of the entropy carried by the size of the generalized influences is harmless, and what must be bounded is the relative entropy of $\mu(S)$ against $\gamma(S)$, which compares a coefficient with the total weight above it. Form (b) replaces the exact coefficients by their noisy upper sets, at any fixed noise level. Form (c) writes B as the expected number of elements a noise-tilted posterior adds to the spectral sample, and shows that only noise levels near $t = 1$ matter. Form (d) reduces the conjecture to the prediction of a single element of the spectral sample.

The equivalences hold for the class of all Boolean functions, not function by function. For instance, when $\hat{f}$ is supported on sets of odd size, as for the majority function, the parity of $|U \setminus \{i\}|$ determines whether $i \in U$, so $E_i(f) = 0$ for every i and $L(f) = 0$, while $H(f)$ may be large compared with $I(f)$.

2. The split

Lemma 2. *For every $T \ne \emptyset$, $\mu(T) \le 4^{|T|-1} \gamma(T)^2$.*

Proof. This is the usual granularity argument for derivatives of Boolean functions [7, Chapter 2]. For $x \in \{-1, 1\}^n$, $\partial_T f(x) = 2^{-|T|} \sum_{y \in \{-1,1\}^T} \big(\prod_{j \in T} y_j\big) f(y, x_{[n]\setminus T})$, a sum of $2^{|T|}$ terms ± 1

divided by $2^{|T|}$. The sum is even, so $\partial_T f$ takes values in $2^{1-|T|}\mathbb{Z}$ and $(\partial_T f)^2 \geq 2^{1-|T|}|\partial_T f|$ pointwise. Hence $\gamma(T) = \mathbb{E}(\partial_T f)^2 \geq 2^{1-|T|}\mathbb{E}|\partial_T f| \geq 2^{1-|T|}|\hat{f}(T)|$. □

Proof of Theorem 1(a). Since $\gamma \leq 1$ and $\gamma \geq \mu$, $A \geq 0$ and $B \geq 0$, so $B \leq H(f)$. By Lemma 2, for $S \neq \emptyset$ in the support, $\log_2(1/\gamma(S)) \leq |S| - 1 + \frac{1}{2}\log_2(1/\mu(S))$, while $S = \emptyset$ contributes 0 to A. So

$$A \leq \sum_{S \neq \emptyset} \mu(S)(|S| - 1) + \tfrac{1}{2}H(f) = I(f) - \mathrm{Var}(f) + \tfrac{1}{2}H(f),$$

and $H(f) = A + B$ gives the claim. □

The constant in Lemma 2 is attained: for a dictator $f = \chi_{\{1\}}$ and $T = \{1\}$ both sides equal 1.

3. Noisy upper sets

Proof of Theorem 1(b). Let $\nu_t(T) = (1-t)^{|T|}\gamma_t(T) = \sum_{U \supseteq T} \mu(U)(1-t)^{|T|}t^{|U \setminus T|}$. This is the law of the random set obtained from U by keeping each element independently with probability $1 - t$, so $\sum_T \nu_t(T) = 1$. Gibbs' inequality gives

$$H(f) \leq \sum_S \mu(S) \log_2 \frac{1}{\nu_t(S)} = A_t + I(f) \log_2 \frac{1}{1-t}.$$

The other bound is $\gamma_t \geq \mu$. □

At $t = 1$ only the lower bound survives: $A \leq H(f)$, and indeed $A \leq H(f)/2 + I(f) - \mathrm{Var}(f)$ by part (a). So the bound $A \leq C\, I(f)$ is a consequence of FEI but is not known to imply it.

4. The tilted excess

Proof of Theorem 1(c). Fix T in the support of μ. Then $\frac{d}{ds}\ln \gamma_s(T) = m_s(T)/s$ for $0 < s \leq 1$, and $m_s(T) = O(s)$ as $s \to 0$. Differentiating once more, $s\frac{d}{ds}m_s(T)$ is the variance of $|U \setminus T|$ under the tilted law in the definition of m_s, so $m_s(T)$ is nondecreasing. Since $\gamma_0(T) = \mu(T)$,

$$\log_2 \frac{\gamma(T)}{\mu(T)} = \frac{1}{\ln 2}\int_0^1 m_s(T)\,\frac{ds}{s}, \qquad \log_2 \frac{1}{\gamma_\lambda(T)} = \log_2 \frac{1}{\gamma(T)} + \frac{1}{\ln 2}\int_\lambda^1 m_s(T)\,\frac{ds}{s}.$$

Averaging the first identity over μ (the integrand is nonnegative) gives the formula for B. Averaging the second gives $A_\lambda = A + \frac{1}{\ln 2}\int_\lambda^1 M_s\, ds/s$. Part (b) at $t = \lambda$ and the bound on A from the proof of part (a) give

$$H(f) \leq \tfrac{1}{2}H(f) + I(f) - \mathrm{Var}(f) + \frac{1}{\ln 2}\int_\lambda^1 M_s(f)\,\frac{ds}{s} + I(f)\log_2 \frac{1}{1-\lambda}.$$

Taking $\lambda = \frac{1}{2}$ gives the first inequality, and $M_s \leq M_1$ gives the second. □

For the equivalence in the theorem, one direction is the inequality just proved, with $ds/s \leq 2\,ds$ on $[\frac{1}{2}, 1]$. For the other, $\int_{1/2}^1 M_s\, ds \leq \int_0^1 M_s\, ds/s = B \ln 2 \leq H(f)\ln 2$.

The quantity $M(f)$ is the expected number of elements that a draw from μ conditioned to contain S adds to S, when S is itself drawn from μ. The bound $H(f) \leq 4I(f) - 2\mathrm{Var}(f) + 2M(f)$ holds for every f; the bound $M(f) \leq C\, I(f)$ for all Boolean f would imply the conjecture, and it is open whether it holds.

5. Predicting one element

Proof of Theorem 1(d). Fix i, and for $T \not\ni i$ write $p = \mu(T)$ and $q = \mu(T \cup \{i\})$. The contribution of the pair to $E_i(f)$ is

$$p \log_2 \frac{p+q}{p} + q \log_2 \frac{p+q}{q}.$$

Summing the second term over i and T gives $I(f)L(f)$. The first term lies in $[0, q/\ln 2]$, and $\sum_i \sum_T q = I(f)$. □

Combined with [10, Corollary 5], which says that FEI with constant C implies $\sum_i E_i \le C\, I(f)$ and that $\sum_i E_i \le C\, I(f)$ for all f implies FEI with constant $C + 2/\ln 2$, part (d) shows that FEI holds if and only if $\sup_f L(f) < \infty$; precisely, $L \le C$ for all f gives $H(f) \le (C + 3/\ln 2) I(f)$.

The loss L has a variational form. For any function $p_i(T) \ge 0$ on pairs, put

$$K_p = \frac{1}{I(f)} \sum_i \sum_{T \not\ni i} \big(\mu(T) + \mu(T \cup \{i\})\big)\, p_i(T).$$

Proposition 3. *For every p, $L(f) \le \log_2 K_p + \mathbb{E}_Q \log_2 \frac{1}{p_i(T)}$, with equality when $p_i(T) = \mu(T \cup \{i\})/(\mu(T) + \mu(T \cup \{i\}))$.*

Proof. $R(T,i) = (\mu(T) + \mu(T \cup \{i\}))\, p_i(T)/(K_p I(f))$ is a probability law on pairs, and Gibbs' inequality $\mathbb{E}_Q \log_2(Q/R) \ge 0$ is the claim. At the stated p, $R = Q$ and $K_p = 1$. □

So a predictor that guesses "$i \in U$" with probability $p_i(T)$, and whose normalized total guess K_p is bounded, bounds L. The constant predictor $p \equiv \frac{1}{2}$ has $K_p = n/(2I(f))$ and gives $L(f) \le \log_2(n/I(f))$, hence $\sum_i E_i(f) \le I(f)\big(\log_2(n/I(f)) + 1/\ln 2\big)$. A bound uniform in n needs a predictor that uses T.

Remark 4. Multiplying f by a character χ_R moves the spectrum, $\widehat{\chi_R f}(S) = \hat{f}(S \triangle R)$. It maps the pairs $\{T, T \cup \{i\}\}$ to pairs, so every E_i is unchanged, while Inf_i becomes $1 - \mathrm{Inf}_i$ for $i \in R$. Choosing $R = \{i : \mathrm{Inf}_i(f) > \frac{1}{2}\}$ shows that the bound $\sum_i E_i(f) \le C\, I(f)$ for all f is equivalent to $\sum_i E_i(f) \le C \sum_i \min(\mathrm{Inf}_i(f), 1 - \mathrm{Inf}_i(f))$ for all f; with [10, Corollary 5], so is FEI. (For H itself the same move is Lemma 2.2 of [11].) Since $D_i f$ takes values in $\{-1, 0, 1\}$, $\mathrm{Inf}_i(f) = \Pr[D_i f \ne 0]$ and $\hat{f}(\{i\}) = \mathbf{E}\, D_i f$, so $\hat{f}(\{i\})^2 \le \mathrm{Inf}_i(f)^2$ and with $W_1(f) = \sum_i \hat{f}(\{i\})^2$,

$$I(f) - W_1(f) \ge \sum_i \mathrm{Inf}_i(f)\big(1 - \mathrm{Inf}_i(f)\big) \ge \tfrac{1}{2} \sum_i \min\big(\mathrm{Inf}_i(f), 1 - \mathrm{Inf}_i(f)\big).$$

Hence FEI is also equivalent to $\sum_i E_i(f) \le C\,(I(f) - W_1(f))$ for all f, with C changed by a factor at most 2.

6. Checks

All identities and inequalities of Theorem 1, Lemma 2, Proposition 3 for the constant predictor and the invariance in Remark 4 were checked with exact Walsh–Hadamard transforms on 119 functions: random functions on 3 to 9 variables with biases $\frac{1}{2}$, 0.15 and 0.03, the AND of eight variables, the tribes functions of widths 2 and 3 on eight and nine variables, and majority on nine variables. The ratio in Lemma 2 has maximum exactly 1; the identities $H = A + B$ and the invariance of $\sum_i E_i$ hold to $2 \cdot 10^{-15}$ and the integral formula for B to $8 \cdot 10^{-7}$ with 80-point Gauss–Legendre quadrature; the inequalities of parts (a) and (b) (at $t = 0.25, 0.5, 0.75, 0.9$) and the bound with $M(f)$ hold on all of these functions, with smallest slack 0.022, 0.0055 and 0.16 respectively (a character has slack 0 in (a) and (b)); and m_s is nondecreasing on the quadrature grid for every function. For majority on nine variables, $\sum_i E_i = L = 0$ while $H/I = 2.137$.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [16] and Math Brain [14]. In November 2026 a Claude Fable agent (Anthropic), connected to the Brain and working alongside Codex agents running OpenAI's GPT-6.1 on the same problem, found the split of part (a), the noisy form of part (b), the integral formula of part (c) and the predictor form of part (d). Claude Opus 5.5 (Anthropic) searched the literature, rederived each proof, found the bound with $M(f)$, wrote the numerical check, prepared the manuscript and audited it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-equivalent-forms`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [14].

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